Evaluation Report

Retrieval, classification, generation quality (DeepEval, RAGAS, LLM-as-Judge) with Claude Sonnet 5.5, prediction models and automated tests

1. Summary

Area	What was measured	Result
Retrieval	30 golden scenarios, Precision@10	keyword 0.36 · semantic 0.50 · hybrid 0.54 · hybrid + auto filters 0.92
Search	180 everyday queries, Precision@10	0.83 (same type and industry), 0.75 with severity ±1; median 60 ms
Classification	30 scenarios, Claude agent	type/industry/region/tier/backup 100%; severity MAE 0.40
DeepEval	10 scenarios	answer relevancy 1.00, faithfulness 1.00, contextual precision 0.87, contextual recall 0.47, recommendation quality 0.56, analysis correctness 0.84
RAGAS	10 scenarios	faithfulness 0.42, faithfulness explanation 0.86, answer relevancy 0.70, context precision 0.65, context recall 0.47
LLM-as-Judge	10 golden scenarios	mean 3.48 / 5, pass rate 60%
LLM-as-Judge (demo set)	10 demo scenarios, full workflow, after plan-writer improvements	mean 3.92 / 5, pass rate 100%
Strategy choice	recommended strategy in the acceptable set	7 of 10 scenarios
Prediction models	20% hold-out	production impact R^2 0.70, full recovery R^2 0.40, high-impact AUC 0.92
Automated tests	pytest, offline	74 tests in 4 files

2. Evaluation Setup

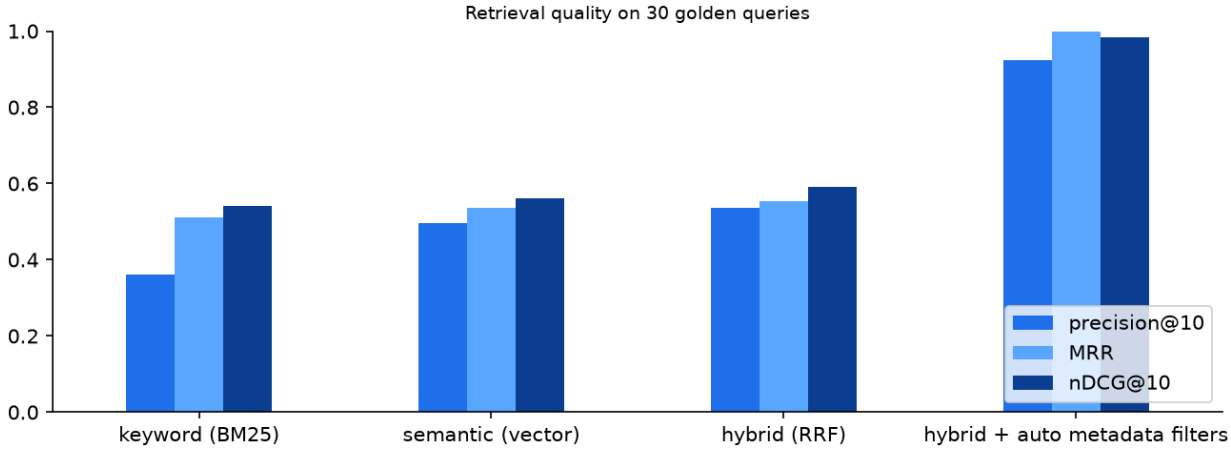
- Golden set: 30 realistic scenarios generated from real dataset records (5 per disruption type, different industries and severities), phrased the way an operations manager would write them, with synonyms and countries instead of dataset labels (evaluation/golden_set.json). Ground truth = the record's attributes.
- Reference answer: the response strategy with the fastest median full recovery among comparable past cases (same type, industry, region, severity band and backup status). Every strategy within 15% of that median is listed as acceptable, because several strategies often recover almost equally fast. Without a backup supplier, Alternative Supplier is never an expected answer (the same feasibility rule the system ranks by, see the Design Document).
- LLM configuration: Claude Sonnet 5.5 (claude-sonnet-5-5) through Anthropic's OpenAI-compatible endpoint generated the answers for the first 10 scenarios (covering all six disruption types) and was also the evaluator for DeepEval, RAGAS and the built-in LLM-as-Judge.
- Metrics that need no LLM (retrieval, search, prediction models, tests) are computed on the full sets.
- The set is synthetic: queries come from templates and the reference comes from the same dataset the system retrieves from, so the scores measure consistency with the historical data, not real-world correctness.

Run of 2026-10-11: Full re-run after the no-backup feasibility fix: retrieval and classification on 30 golden scenarios, then 10 scenarios answered and judged by Claude Sonnet 5.5, scored with DeepEval and RAGAS. Every answer, plan and judgement came from Claude (no template or checklist fallbacks). Changes since the previous run: Alternative Supplier is ranked last and never recommended when there is no backup supplier; the golden set applies the same rule (10 of 30 acceptable-answer sets changed). Token limits raised (plan 4000, judge 2500, classifier 1200) because Claude's replies were being cut off mid-JSON. The LLM judge scored this run lower than the 2026-10-09 run (mean 3.48 vs 4.12). Three of the four failing scenarios have a backup supplier, so their ranking did not change; the judge's reasons are about missing owners/deadlines and the ranking table's blended recovery figure differing from the plan's model figure.

3. Retrieval Results

3.1 Golden scenarios (30 queries, relevance: same type & industry, severity +/-1)

Retriever	precision@10	hit@1	hit@10	MRR	nDCG@10
keyword (BM25)	0.360	0.400	0.767	0.511	0.540
semantic (vector)	0.497	0.500	0.633	0.536	0.560
hybrid (RRF)	0.537	0.500	0.667	0.554	0.591
hybrid + auto metadata filters	0.923	1.000	1.000	1.000	0.984



Findings: on long, paraphrased scenarios BM25 alone is weak (it cannot map 'hackers' to Cyber Attack) and semantic search carries most of the signal; unfiltered hybrid fusion is slightly better than semantic alone. The largest gain comes from metadata filters derived by the classification agent: Precision@10 rises to 0.92 and MRR to 1.00. This is the configuration the multi-agent workflow uses.

Sensitivity to the semantic weight α (hybrid, no filters)

α	precision@10	hit@1	hit@10	MRR	nDCG@10
0.3	0.363	0.433	0.767	0.540	0.568
0.5	0.483	0.500	0.933	0.643	0.696
0.6	0.537	0.500	0.667	0.554	0.591
0.7	0.530	0.500	0.667	0.554	0.591
0.8	0.520	0.467	0.667	0.530	0.577
0.9	0.503	0.500	0.667	0.540	0.570

α = 0.5 gives the best ranking (MRR, nDCG, Hit@10) and α ≥ 0.6 the best Precision@10; the default α = 0.6 favours a precise top-10 list and is adjustable per request. The sweep reuses the same 30 queries, so it is indicative rather than a held-out tuning result.

Example retrievals (hybrid + auto filters)

Query: Hackers breached the erp of our Tier-1 medium semiconductor supplier in Vietnam. Output is significantly disrupted. We have no backup supplier. What happened in similar cases and how should we respond?

Case	Type	Industry	Severity	Score
DIS-030427	Cyber Attack	Electronics	3	0.601
DIS-092452	Cyber Attack	Electronics	3	0.600
DIS-069940	Cyber Attack	Electronics	3	0.590

Query: An equipment breakdown stopped the production line at our Tier-3 small pharma active ingredient supplier in Taiwan. There are moderate delays. We have no backup supplier. What happened in similar cases and how should we respond?

Case	Type	Industry	Severity	Score
DIS-097820	Factory Incident	Pharmaceuticals	3	0.726
DIS-072166	Factory Incident	Pharmaceuticals	3	0.674
DIS-087565	Factory Incident	Pharmaceuticals	3	0.644

3.2 Everyday search queries (180 queries)

The Similar-cases page gets short, loose queries rather than long scenarios, so a second set was built: 6 disruption types x 5 industries x 6 phrasings (keywords, keywords + place, short sentence, full sentence, with severity...). Relevance: loose = same type and industry; strict = also severity within +/-1 (only queries that state a severity).

Method	P@10 loose	P@10 strict	Hit@10 loose	MRR loose	nDCG@10 loose	Median ms
keyword	0.532	0.417	0.906	0.633	0.696	39
semantic	0.833	0.748	0.850	0.837	0.841	60
hybrid a=0.4	0.607	0.470	0.956	0.772	0.800	68
hybrid a=0.5	0.737	0.630	0.972	0.898	0.885	66
hybrid a=0.6 (default)	0.834	0.748	0.850	0.843	0.844	60
hybrid a=0.7	0.835	0.755	0.850	0.843	0.844	64

Filters	Median ms	P95 ms	Median matching records	Results breaking a filter
1 filter (type)	107	146	15,016	0
2 filters (type + industry)	76	90	3,092	0
4 filters (+ severity band, region)	80	98	1,601	0

Findings: on short queries hybrid search ties semantic search (it does not beat it on this set) and both are far ahead of keyword search; α 0.6-0.7 is best. Filtered searches use an exact search over the vectors in memory and take well under 0.2 s, and no result ever broke a filter.

3.3 Known retrieval bias: the word 'supplier'

Every narrative ends with the response taken (e.g. 'Response: Alternative Supplier initiated after...'), so a query containing the word 'supplier' can match that label. A paired test searched each of 150 queries twice, as written and with the word removed: the share of Alternative Supplier cases in the top 10 was 54% as written vs 34% without the word (difference 20%, 95% CI 16%-23%; dataset share 44%). The strategy recommendation is not affected, because strategies are compared on a pool of comparable cases selected by profile, not by wording. A fix (embed only the situation part of each narrative) needs an index rebuild and is listed as future work.

4. Classification Agent

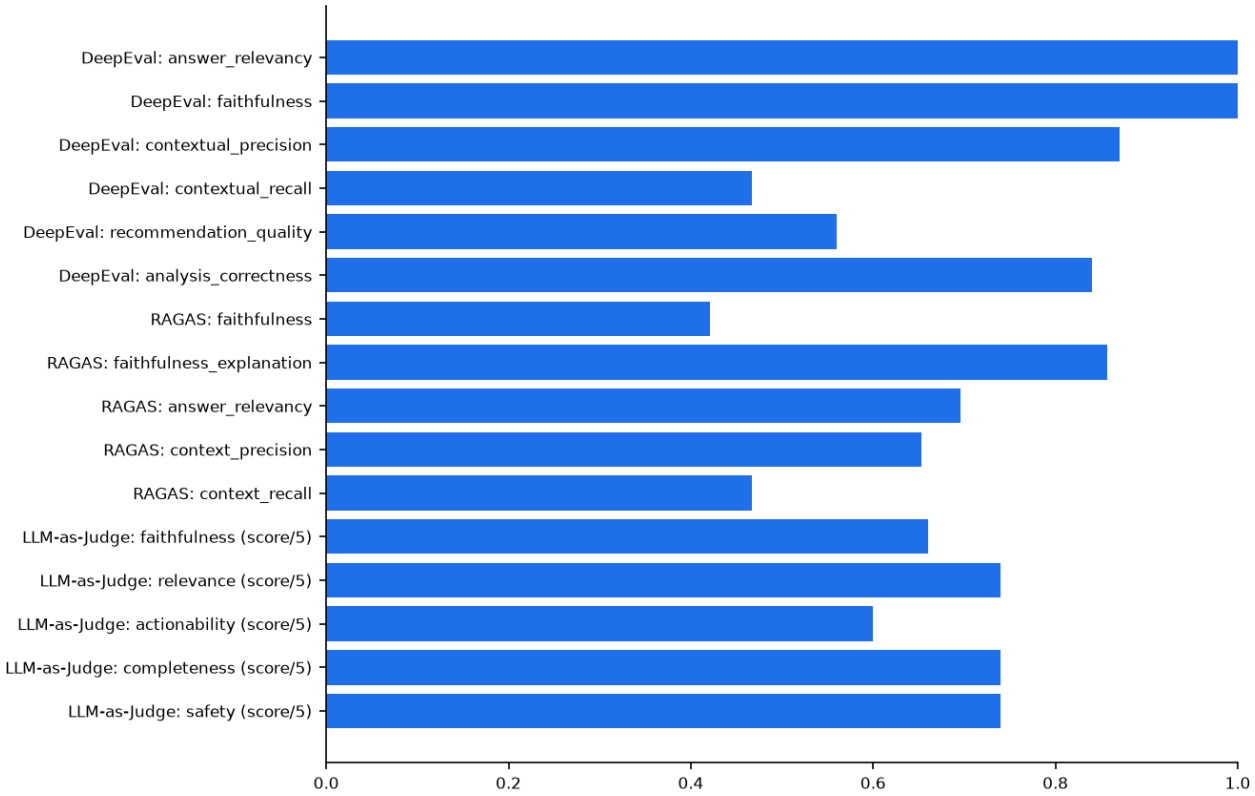
Engine	disruption_type	industry	supplier_region	supplier_tier	has_backup_supplier	severity MAE	severity ±1
Rule engine (offline fallback)	1.000	1.000	1.000	1.000	1.000	0.100	0.967
Claude Sonnet 5.5 agent	1.000	1.000	1.000	1.000	1.000	0.400	1.000

Both engines extracted every categorical attribute correctly on all 30 scenarios. Severity is the hardest attribute because it is inferred from wording such as 'halted for weeks': the Claude agent's mean error was 0.40 levels (100% within one level), the rule engine's 0.10 (97% within one level). The rule engine is the

offline fallback; on these template-phrased scenarios its fixed severity phrases match exactly, which is why it does so well here.

5. Generation Quality (DeepEval, RAGAS, LLM-as-Judge)

Framework	Metric	Mean (0-1)	n
DeepEval	answer_relevancy	1.000	10
DeepEval	faithfulness	1.000	10
DeepEval	contextual_precision	0.870	10
DeepEval	contextual_recall	0.467	10
DeepEval	recommendation_quality	0.560	10
DeepEval	analysis_correctness	0.840	10
RAGAS	faithfulness	0.421	10
RAGAS	faithfulness_explanation	0.857	10
RAGAS	answer_relevancy	0.696	10
RAGAS	context_precision	0.653	10
RAGAS	context_recall	0.467	10
LLM-as-Judge	faithfulness (score/5)	0.660	10
LLM-as-Judge	relevance (score/5)	0.740	10
LLM-as-Judge	actionability (score/5)	0.600	10
LLM-as-Judge	completeness (score/5)	0.740	10
LLM-as-Judge	safety (score/5)	0.740	10



LLM-as-Judge: mean 3.48 / 5 and a 60% pass rate (pass = mean ≥ 3.5 and faithfulness ≥ 3). The judge prompt asks for strict scoring (5 only when nothing can be improved). The failing plans lost points for missing owners and deadlines, for not flagging a close call between the top two strategies, and because the ranking table's blended recovery figure (history + model) differs from the model figure the plan quotes. None failed for

recommending a backup supplier the user does not have.

Reading the scores

- Strong: DeepEval answer relevancy (1.00), faithfulness (1.00) and contextual precision (0.87), and RAGAS explanation faithfulness (0.86): the plans answer the question and stick to the evidence they were given.
- Whole-plan faithfulness in RAGAS (0.42) vs explanation-only (0.86): RAGAS checks every sentence as a factual claim, but action steps ('open a daily call with the supplier') are recommendations that no evidence can support, so they count as unsupported. The explanation paragraph is the part that makes factual claims.
- Contextual recall (DeepEval 0.47, RAGAS 0.47) and recommendation quality (0.56) are measured against the reference answer, which is computed from the true record. When the classified severity differs from the record's (for example 4 instead of 3), the system compares strategies on a different pool of cases, so the reference's figures are not in its evidence and a different strategy can win.
- RAGAS answer relevancy (0.70) vs DeepEval (1.00): RAGAS generates questions from the answer and compares them to the original question by embedding similarity; long, multi-part plans score lower on that measure.
- Self-evaluation: Claude both wrote and evaluated the answers, which can favour its own style. The label-based metrics (retrieval, classification, strategy agreement, models) do not depend on an LLM and act as a check.

Per-scenario results

Scenario	Type	Severity (true / classified)	Recommended	Acceptable	Judge	Latency
DIS-065446	Cyber Attack	3 / 3	Inventory Buffer	yes: Inventory Buffer	3.8 PASS	20 s
DIS-088555	Factory Incident	2 / 3	Inventory Buffer	yes: Customer Delay, Inventory Buffer, Production Reroute	3.4 FAIL	22 s
DIS-092208	Geopolitical	3 / 3	Alternative Supplier	no: Inventory Buffer	3.6 PASS	20 s
DIS-022328	Labor Strike	1 / 1	Production Reroute	no: Inventory Buffer	3.6 PASS	19 s
DIS-022526	Natural Disaster	3 / 3	Inventory Buffer	yes: Inventory Buffer	3.6 PASS	19 s
DIS-062333	Port Congestion	2 / 3	Alternative Supplier	yes: Alternative Supplier	3.0 FAIL	19 s
DIS-057044	Cyber Attack	3 / 3	Alternative Supplier	yes: Combined Strategy, Alternative Supplier, Production Reroute	3.2 FAIL	18 s
DIS-029375	Factory Incident	2 / 3	Inventory Buffer	no: Combined Strategy	3.8 PASS	19 s
DIS-013272	Geopolitical	2 / 3	Inventory Buffer	yes: Combined Strategy, Production Reroute, Inventory Buffer	3.6 PASS	18 s
DIS-096506	Labor Strike	3 / 3	Alternative Supplier	yes: Production Reroute, Alternative Supplier, Inventory Buffer	3.2 FAIL	18 s

Strategy agreement: the recommended strategy was in the acceptable set in 7 of 10 scenarios. The strategy itself is chosen by the scoring system (outcomes of comparable past cases, ML what-if predictions and feedback), not by the LLM. Of the 3 misses, 1 followed a severity misclassification (DIS-029375), which changes the comparable pool. In the others the severity was right, but the reference looks only at the fastest

median recovery, while the score also weighs revenue loss, supplier-change risk, the amount of evidence and the ML estimate for this profile, so a different strategy came out on top. Average latency per full analysis with Claude: 19 s.

6. Prediction & Risk Models (20% hold-out)

Target	MAE	RMSE	R^2	Baseline MAE	P10-P90 coverage
production_impact_pct	7.80	9.70	0.699	14.29	0.793
revenue_loss_usd	1586665.64	4535647.69	0.333	2210329.11	0.785
full_recovery_days	31.00	44.16	0.399	40.72	0.791
partial_recovery_days	15.28	22.04	0.272	17.48	0.782

Classifier	ROC-AUC	PR-AUC	Threshold	Precision	Recall	F1	Positive rate
high_impact	0.924	0.819	0.39	0.712	0.750	0.730	0.253
permanent_supplier_change	0.650	0.358	0.19	0.300	0.755	0.429	0.240

Model benchmark (same split; is the model or the data the limit?)

Target	Median baseline	LightGBM	Random forest	Linear
production_impact_pct (R^2)	0.00	0.699	0.699	0.678
revenue_loss_usd (R^2)	0.00	0.333	0.328	0.331
full_recovery_days (R^2)	0.00	0.399	0.403	0.394
partial_recovery_days (R^2)	0.00	0.272	0.275	0.269
high_impact (ROC-AUC)	0.500	0.924	0.923	0.925
permanent_supplier_change (ROC-AUC)	0.500	0.650	0.652	0.657

All regressors beat the median baseline. Production impact is the most predictable outcome (driven mainly by severity); revenue loss and recovery time carry more noise. Three very different model families score the same, so the limit is the information in the dataset's features, not the model. The P10-P90 ranges contain the true value about 79% of the time (target 80%). The supplier-change classifier is weak (AUC 0.65); its threshold is tuned for F1 and it is reported as a probability, not a decision.

7. Automated Test Cases

Test file	Tests	Covers
test_api.py	24	endpoint contracts, validation and guardrail rejections, streaming, feedback loop, deep check, judge fallback reasons, risk typical-severity view, minor-event note
test_core.py	12	rule classifier, prioritization, hybrid search filters and modes, prediction ranges, strategy ranking (strategy is never the LLM's choice), unsupported-number check
test_guardrails.py	8	prompt injection, off-topic detection, length limits, PII masking, hallucinated citation removal
test_investigation.py	30	retrieval query cleaning, severity band and relaxation, comparable-case pool, fact sheet, plan guardrails, override and ambiguity notes, severity from durations, region mapping

The suite runs fully offline (LLM disabled, throwaway feedback database): python -m pytest -q

```
    warnings.warn(
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
74 passed, 11 warnings in 97.70s (0:01:37)
```

8. Demo Scenarios (full workflow, after plan-writer improvements)

Ten further scenarios, each written from a real dataset record but phrased with the mixed signals a real report has (stock runway, a backup that only covers part of the volume, a restart already under way, a customer deadline). Together they cover all six disruption types, every industry and region twice, severities 1-5, and five cases with a backup supplier and five without. Each ran through the same path as the app (all agents, the judge and its revision loop) on Claude Sonnet 5.5, after the plan writer was given the priority level, a minor-event note and owner/deadline/trigger rules. The golden-set results above were measured before these improvements.

#	Scenario	Priority	Recommended	Judge	Verdict
1	Ice storm at a deep-tier substrate supplier	P4	Production Reroute	4.0	PASS
2	Ransomware at a sole-source brake supplier	P2	Inventory Buffer	3.6	PASS
3	Strike at an active-ingredient supplier	P2	Inventory Buffer	3.8	PASS
4	Rotterdam congestion holding aerospace forgings	P3	Inventory Buffer	4.0	PASS
5	Export restrictions block packaging from Egypt	P2	Inventory Buffer	4.0	PASS
6	Fire destroys an aerospace fastener plant	P1	Combined Strategy	4.0	PASS
7	Floods cut off a sterile vial supplier	P2	Alternative Supplier	4.0	PASS
8	Malware outage at a board contract manufacturer	P3	Inventory Buffer	3.6	PASS
9	One-day walkout at a harness supplier	P4	Combined Strategy	4.2	PASS
10	Hamburg backlog delays packaging before a launch	P4	Combined Strategy	4.0	PASS

Mean judge score 3.92 / 5; 10 of 10 pass. Scores move by up to about 0.4 between runs because the LLM's wording changes. Scenario texts: evaluation/demo_scenarios.json; reproduce with python -m evaluation.demo_scenarios.

9. Sample Outputs (Claude Sonnet 5.5)

Scenario DIS-065446

Query: Hackers breached the erp of our Tier-1 medium semiconductor supplier in Vietnam. Output is significantly disrupted. We have no backup supplier. What happened in similar cases and how should we respond?
Expected: Cyber Attack / Electronics / severity 3 · Classified: Cyber Attack / Electronics / severity 3
Reference: This is a Cyber Attack disruption (severity 3) affecting a Tier-1 Electronics supplier in Asia-Pacific, no backup supplier. Among 706 comparable past cases, Inventory Buffer had the fastest median full recovery (80 days) among the options available without a backup supplier. Recommended: Inventory Buffer.

```
Recommended strategy: Inventory Buffer.
Rationale: Inventory Buffer is the chosen response because it has the fastest model-estimated partial recovery of the
    options (~28 days vs ~47 days for the Customer Delay fallback) and a full recovery of ~97 days vs ~117 days for th
    e fallback. It has also been used in 106 of 706 comparable past cases, where the median full recovery was 80 days,
     and there is no backup supplier to switch to. The trade-off is a 30% risk of a permanent supplier change and an e
    stimated revenue loss of ~$1.4M.
Immediate actions: Quantify on-hand and in-transit semiconductor inventory and calculate how many days of production i
    t covers against the ~28-day partial recovery window.; Prioritize buffer stock for the highest-margin and most cri
    tical product lines, and set allocation rules so the buffer lasts as long as possible.; Open a direct line with th
    e Vietnam supplier to get a status on ERP restoration, what output is possible manually, and expected shipment dat
    es.; Alert key customers early about possible constrained supply, and prepare Customer Delay as the fallback if th
    e buffer runs short.; Start qualifying a new semiconductor source as a parallel, long-lead-time step; it is not an
     immediate option and should not be counted on for the near term.
Recovery plan: Days 0-28: Draw down the buffer under the allocation rules, monitor consumption daily, and track the su
    pplier's ERP restoration progress toward the expected ~28-day partial recovery.; Around day 28: Confirm partial su
    pplier output has resumed, verify shipment integrity and order data given the breach, and begin rebuilding buffer
    stock at a controlled pace.; Days 28-97: Ramp supplier volumes toward full capacity and keep reviewing buffer leve
    ls; if stock is projected to run out before supply stabilizes, switch to the Customer Delay fallback (~117 days fu
    ll recovery).; By ~97 days: Confirm full recovery, restore normal inventory targets, and review whether the new-so
    urce qualification should continue given the 30% permanent supplier-change risk.
Risks: The buffer may run out before the supplier recovers: the model expects ~97 days to full recovery, while the pas
    t-case median for this strategy is 80 days, and the closest examples with this strategy ranged from 48 days (DIS-0
    35342) to 171 days (DIS-038405).; With no backup supplier, a longer-than-expected outage leaves no immediate way t
    o source replacement supply, and the Customer Delay fallback is slower (~117 days full recovery).; There is a 30%
    risk of a permanent supplier change, and 88% of the 8 similar historical disruptions ended with one, so supplier r
    elationship and sourcing exposure may change even if the response works.
```

Judge: 3.8/5 (PASS) · latency 20.34s

Scenario DIS-088555

Query: An equipment breakdown stopped the production line at our Tier-3 small pharma active ingredient supplier in Taiwan. There are moderate delays. We have no backup supplier. What happened in similar cases and how should we respond?
Expected: Factory Incident / Pharmaceuticals / severity 2 · Classified: Factory Incident / Pharmaceuticals / severity 3
Reference: This is a Factory Incident disruption (severity 2) affecting a Tier-3 Pharmaceuticals supplier in Asia-Pacific, no backup supplier. Among 318 comparable past cases, Customer Delay had the fastest median full recovery (62 days) among the options available without a backup supplier. Recommended: Customer Delay; Inventory Buffer, Production Reroute would also be acceptable (within 15% of the best median recovery).

```
Recommended strategy: Inventory Buffer.
Rationale: Inventory Buffer is the chosen response: it was used in 56 of the 277 comparable cases, and the model estim
    ates partial recovery in ~33 days and full recovery in ~107 days. That is faster than the Production Reroute fallb
    ack (~42 days partial, ~119 days full) and carries a lower permanent supplier-change risk (24% vs 27%). Because th
    ere is no backup supplier, bridging the gap with stock already on hand is the most practical way to protect supply
     while the Taiwan line is repaired. The estimated revenue loss is ~$506K.
Immediate actions: Count all active ingredient stock on hand, in transit and at downstream sites, and calculate how ma
    ny days of supply it covers against the ~33-day partial recovery and ~107-day full recovery estimates.; Prioritize
     the buffer for the highest-criticality products and customers, and ration allocation so stock lasts through the e
    xpected recovery window.; Contact the Tier-3 supplier for a repair plan: root cause of the breakdown, parts and te
    chnician availability, restart date, and whether a partial restart is possible.; Place the largest practical order
    s for the output the supplier can still ship, and ask for batches to be released as they are completed rather than
     held for a full lot.; Start qualifying a new source in parallel as a long-lead-time step only. It is not a near-t
    erm option, especially for a regulated input, and does not replace the buffer.
Recovery plan: Days 0-14: Confirm buffer coverage, set allocation rules, and get the supplier's repair schedule and re
    start date. Report the coverage position to operations and commercial teams weekly.; Days 15-33: Track repair prog
    ress against the supplier's plan. Expect partial recovery around day 33 and begin receiving reduced volumes. Use t
    hese receipts to slow buffer drawdown rather than relax allocation.; Days 34-107: Ramp supplier output toward norm
    al while rebuilding the buffer gradually. Keep allocation limits until volumes are stable, and expect full recover
    y around day 107.; Throughout: Review the position at ~33 days. If partial recovery has not started, consider movi
    ng to the Production Reroute fallback (~42 days partial, ~119 days full).
Risks: The buffer may run out before full recovery at ~107 days. Past cases using this strategy varied widely, from 63
     days (DIS-075871) and 85 days (DIS-097820) to 202 days (DIS-065219), so a longer outage is possible.; There is a
    24% risk of a permanent supplier change, and 50% of similar past disruptions ended that way. With no backup suppli
    er in place, a lengthy outage could force a difficult, slow requalification.; Drawing down the buffer leaves less
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    cover against any further disruption until stock is rebuilt, and the estimated revenue loss of ~$506K could rise i
    f shortages reach customers.
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Judge: 3.4/5 (FAIL) · latency 21.68s

Scenario DIS-092208

Query: New export sanctions have blocked our Tier-1 large consumer goods packaging supplier in Canada. Output is significantly disrupted. We have a backup supplier. What happened in similar cases and how should we respond?
Expected: Geopolitical / Consumer Goods / severity 3 · Classified: Geopolitical / Consumer Goods / severity 3
Reference: This is a Geopolitical disruption (severity 3) affecting a Tier-1 Consumer Goods supplier in North America, with a backup supplier. Among 352 comparable past cases, Inventory Buffer had the fastest median full recovery (49 days).
Recommended: Inventory Buffer.

```
Recommended strategy: Alternative Supplier.
Rationale: Alternative Supplier is the chosen strategy and fits because a backup supplier is already in place, so the
    blocked Canadian Tier-1 source can be bypassed rather than waited out. It was used in 245 of 352 comparable past c
    ases (median full recovery 58 days), and the model estimates ~22 days to partial recovery and ~67 days to full rec
    overy for this disruption, against an estimated revenue loss of ~$2.3M. The fallback, Inventory Buffer, is estimat
    ed slightly slower at ~69 days to full recovery.
Immediate actions: Confirm the backup supplier's available capacity, lead times and ability to meet packaging specific
    ations, and place initial orders to start moving volume away from the sanctioned Canadian supplier.; Have legal an
    d compliance review the new export sanctions to confirm which in-transit and on-order materials from the blocked s
    upplier can still be received, and which cannot.; Take stock of on-hand packaging inventory and open orders, and p
    rioritise the highest-revenue product lines for the first backup-supplier volumes, given the ~$2.3M estimated reve
    nue loss.; Brief production planning and key customers on expected output constraints during the ~22-day ramp to p
    artial recovery.; Prepare the Inventory Buffer fallback by identifying which packaging stock could be held back or
     reallocated if the backup supplier ramps up slower than planned.
Recovery plan: Days 0-22: Onboard the backup supplier, run quality checks on first deliveries and ramp volumes toward
    partial recovery (~22 days).; Days 22-45: Scale backup supplier volumes and restore the remaining product lines, r
    eviewing weekly against the expected recovery path.; Days 45-67: Stabilise supply and close the remaining gaps, wo
    rking toward full recovery in ~67 days.; After full recovery: Review sourcing arrangements, since 38% of similar h
    istorical disruptions ended with a permanent supplier change, and decide whether to formalise the backup supplier'
    s role.
Risks: The backup supplier may not ramp as fast as planned, which would push recovery past the ~67-day estimate; the I
    nventory Buffer fallback (~69 days) would then come into play.; There is a 21% risk of a permanent supplier change
    , and 38% of similar historical cases ended that way, which could lock in new supplier terms or costs.; Past outco
    mes vary widely: the Alternative Supplier cases cited ranged from 23 to 68 days to full recovery, so the timeline
    is uncertain and revenue loss could differ from the ~$2.3M estimate (similar cases ranged from $0.64M to $5.54M).
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Judge: 3.6/5 (PASS) · latency 19.63s